Spatial Localization Microphone Array

Working Technical Report

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Introduction

The spatial localization project uses a four-microphone square array to actively track a human speaker. Using omnidirectional microphones, preamplifier circuits, an analog-to-digital converter (ADC), and a microcontroller (MCU), the system is designed to locate a speaker within a room with high precision using time difference of arrival (TDoA). This report is an ongoing document that tracks the design decisions, research, and future steps for the project. It covers the specific parts and circuit layouts, the reasoning behind them, and whether they are subject to change.

Method

Shape of the Array

Selecting the shape of the array was the most fundamental decision in the project, because TDoA analysis is sensitive to the spacing and relative location of the microphones. Many options were available, but the research pointed to the square array as the best option. The first decision was between a coplanar and a non-coplanar array. Since the horizontal angle of the speaker is the most important output, the added complexity of non-coplanar microphones was avoided [1]. Next, the shape of the coplanar design was selected. Two microphones would provide the angle but would have a symmetrical ambiguity about the microphone axis. A three-microphone system would solve this, but four-microphone systems offer higher position precision [2] without greatly increasing the mathematical or geometric complexity.

The spacing of the microphones was the next decision. Human speech carries the majority of its information in the 300–3400 Hz range [3]. Using 3400 Hz as the upper frequency limit, the size of the array could be established. The wavelength can be calculated using Equation 1.

$$\lambda = v/f$$

Equation 1.

Using 3400 Hz as the upper limit and a speed of sound of 343.14 m/s in a 20 °C room [4], a λ of 0.101 m was found. The Nyquist theorem states that, to prevent spatial aliasing, the largest distance in the array should be half this length, about 0.05 m (5 cm). Using this as

the corner-to-corner (diagonal) length gives a side length of $5/\sqrt{2} \approx 3.54$ cm for the final design.

The Microphone Circuit

Calculating Gain

The first part of the project was to design a circuit that could properly collect and amplify the sound signal. An omnidirectional microphone was selected for this project; it only converts the pressure of the sound wave into an electrical signal. The specific microphone is likely to change depending on available parts. The first drafts of the circuits were made using the Infineon IM73A135V01. This is a versatile, high-range analog microphone with a sensitivity (S) of 12.6 mV/Pa rms, or −38 dBV [5]. The typical output of the microphone due to a human speaker was then calculated. Rearranging the formula for sound pressure level in dB SPL (Equation 2) gives the formula for pressure (Equation 3).

$$dB_{SPL} = 20log_{10}(\frac{p}{p_0})$$

Equation 2.

$$p = p_0 10^{dB_{SPL}/20}$$

Equation 3.

Assuming that a normal voice is around 60 dB SPL when it reaches the microphone, the output voltage was calculated using Equation 4.

$$V_{out} = S \cdot p_{input}$$

Equation 4.

The pressure at 60 dB SPL was calculated to be 0.02 Pa. Using this and the sensitivity (S) of 12.6 mV/Pa gives a voltage output of 252 μV rms. This signal needs to be amplified to fit the range of the ADC as closely as possible. The selected ADC is the Texas Instruments ADS131M04, chosen for its simultaneous four-channel sampling, which avoids any sampling skew between channels and microphones. Additionally, its analog supply can operate at 2.8 V, which is the main power rail voltage for the analog circuitry. The full differential range of the ADC is ±1.2 V. To avoid clipping, the total gain from microphone to digital signal needs to keep the signal within that range. A study on the maximum dB SPL measured at 1 m in “normal” speaking conditions found a value of 79.6 dB. Using this as the maximum expected signal, an optimal system gain could be found. Using that 79.6 dB

value in Equations 3 and 4, the expected voltage output is 2.4 mV rms. Converting 2.4 mV rms to peak-to-peak voltage uses Equation 5.

$$V_{pp} = 2\sqrt{2}V_{rms}$$

Equation 5.

This results in a value of 6.8 mVpp. A gain of 352 would completely fill the dynamic range of the ADS131M04. Filling this range is unnecessary: too much gain will increase noise in the signal, and the ADC can still work perfectly well without filling it. A smaller preamp gain of 21 was selected for the microphone op-amp circuit, combined with a 4× PGA gain in the ADC, for a total gain of 84×. This gives an expected final signal of 571 mVpp at 79.6 dB SPL.

Calculating the High-Pass and Low-Pass Filters

To simplify the signal, high-pass and low-pass filters were added to the circuit, using the 300–3400 Hz range of important human speech as the limits. First, the high-pass filter needed to pass frequencies above 300 Hz; the required capacitance was calculated using Equation 6.

$$C = \frac{1}{2\pi R f_c}$$

Equation 6.

Using a high resistance of 100 kΩ, this calculation gives a capacitance of 5.31 nF. Since 5.31 nF is not a standard value, a 5.6 nF capacitor was selected, resulting in a high-pass cutoff of 284 Hz. The high-pass circuit is shown in Figure 1 below.

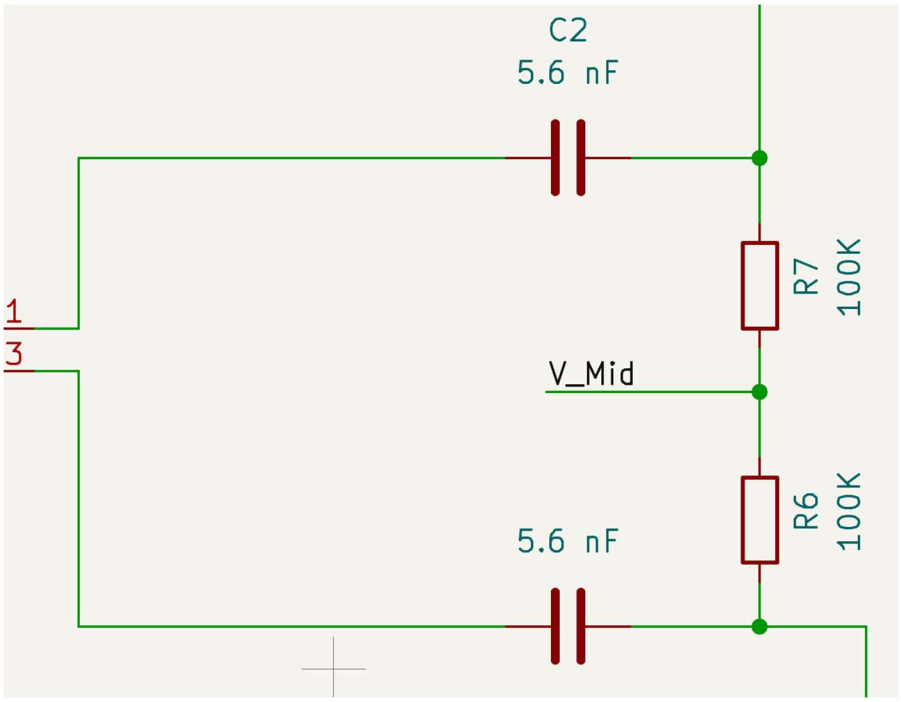


Figure 1. High-pass filter

Next, the low-pass filter was designed. The filter was placed in the feedback section of the op-amp to simplify the circuit, with the low-pass capacitors in parallel with the feedback resistors. To achieve the aforementioned gain of 21, 10 kΩ feedback resistors were selected. Using this value and the low-pass cutoff frequency of 3400 Hz, the optimal capacitance was calculated using Equation 6, giving a value of 4.68 nF. Using the standard capacitance of 4.7 nF, the cutoff frequency was calculated to be 3.39 kHz. The low-pass filter is shown in Figure 2 below.

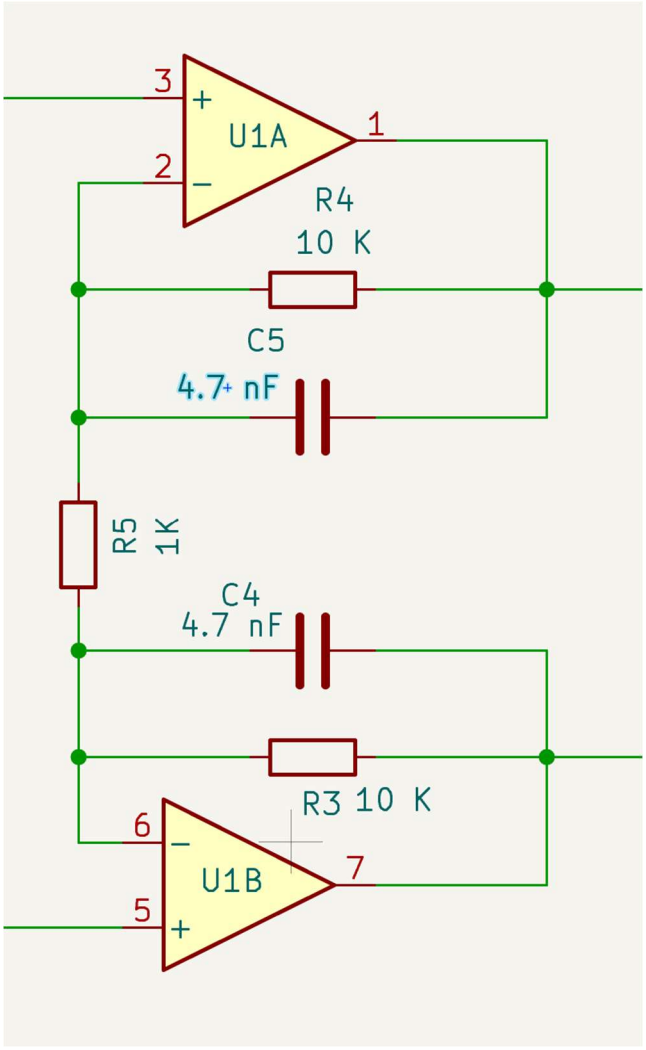


Figure 2. Low-pass filter

The grounds on the op-amps are connected so that the amplification stays consistent between the two; the negative and positive microphone outputs should be amplified by the same amount to keep the signal consistent.

The ADC Circuit

Calculating Circuit Elements

First, resistors needed to be placed after the op-amp outputs to isolate the ADC from the preamp system. This also prevents any spikes from the ADC from feeding back into the op-amps. The concern is that these resistors create a voltage divider with the input impedance of the ADC. According to the datasheet, the input impedance is 330 kΩ [6]. The voltage divider formula is shown in Equation 7 below, where $R_{selected}$ is the resistor placed after the op-amp.

$$V = \frac{R_{ADC}}{R_{ADC} + R_{selected}}$$

Equation 7. Voltage divider

To keep this loss down, a 1 kΩ resistor was selected. Putting this into Equation 7 results in a voltage ratio of 0.994, so very little loss takes place with a 1 kΩ resistor.

Next, certain frequencies needed to be filtered out. The selected ADC has a sampling frequency of 4.096 MHz and an output data rate of 32 kSPS (kilosamples per second) in low-power mode [6]. Additionally, the ADC has a digital filter that removes signals at and above 16 kHz, the Nyquist frequency of the 32 kSPS output. The concern is that this filter will not remove signals close to the sampling frequency of the ADC, as these signals would alias directly into the audio band. An input RC filter using the 1 kΩ resistor selected above was designed to remove these signals, using the cutoff frequency equation (Equation 6). To smooth out the signal for the ADC, the filter capacitance needed to be much larger than the ADC's internal sampling capacitance of 0.74 to 1.3 pF [6]. Aiming for a cutoff of 32 kHz, the optimal capacitance was calculated to be 2.48 nF. The nearby standard value of 2.2 nF was selected, which gives a cutoff of 36 kHz, plenty for this purpose.

After this input RC filter, the signal is ready to be digitally processed. The schematic is shown below in Figure 3.

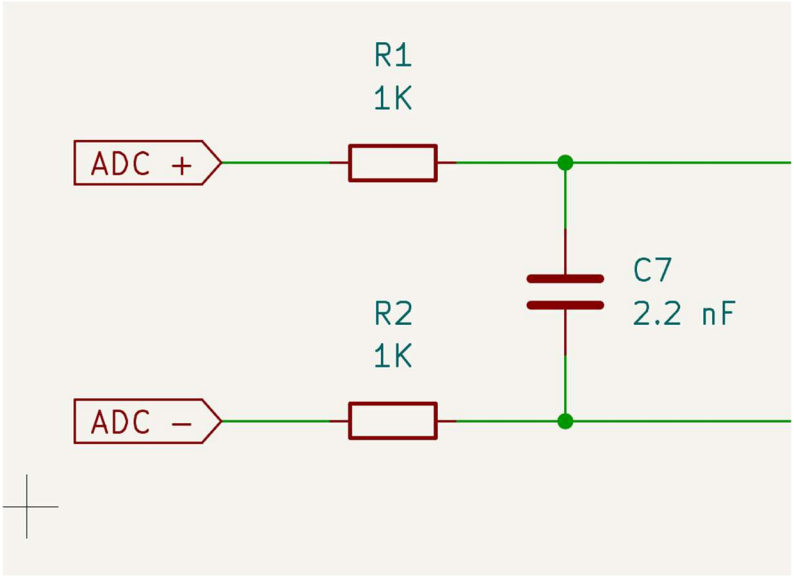


Figure 3. Input RC filter

This input RC filter was repeated for all four microphones. Figure 4 below shows the four input RC filters and their connections to the ADC.

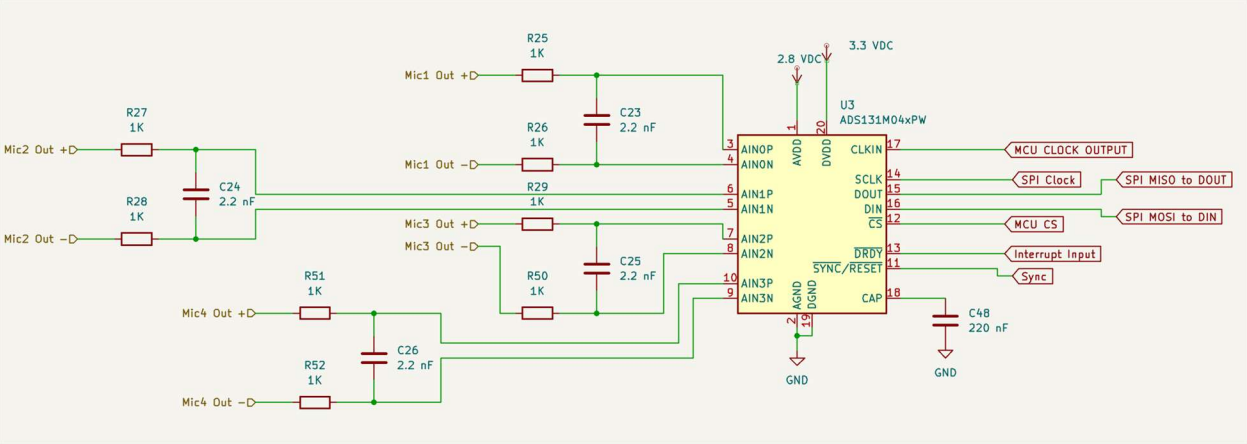


Figure 4. ADC circuit

Figure 4 also shows the connections to the MCU, which are described in the next section.

The MCU

For the MCU, the ESP32-S2-WROVER was selected. Its relatively high clock speed of up to 240 MHz [7] will allow for fast processing of the ADC data. The primary motivator in the selection was availability, as the module is readily available for free to Northeastern students.

The ESP32-S2 has a supply voltage requirement of 3.0 to 3.6 V [7], and the ADC's DVDD supply has a range of 2.7 to 3.6 V [6]. Because the ESP32-S2 cannot run from the 2.8 V rail used by the rest of the circuit, a secondary 3.3 V supply was chosen for the ESP32-S2 and DVDD, as it falls well within both voltage ranges and is a standard value.

The ADC requires several connections to the ESP32-S2, and all ADC pin names correspond to those in Figure 4 above. These include a clock input at CLKIN (pin 17), an SPI clock at SCLK (pin 14), a master-in slave-out (MISO) line that sends data to the MCU from DOUT (pin 15), a master-out slave-in (MOSI) line that carries data from the MCU to DIN (pin 16), a chip select at CS (pin 12), a data-ready interrupt at DRDY (pin 13), and finally a sync pin, which is unnecessary for this project.

The ESP32-S2's pin assignments are very flexible: each of the connections to the ADC can be mapped to any GPIO pin on the MCU [7]. The module pinout is shown below in Figure 5.

Name	No.	Type[1]	Function
GND	1	P	Ground
3V3	2	P	Power supply
IO0	3	I/O/T	RTC_GPIO0, GPIO0
IO1	4	I/O/T	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0
IO2	5	I/O/T	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
IO3	6	I/O/T	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
IO4	7	I/O/T	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
IO5	8	I/O/T	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
IO6	9	I/O/T	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5
IO7	10	I/O/T	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
IO8	11	I/O/T	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7
IO9	12	I/O/T	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD
IO10	13	I/O/T	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICS0, FSPIIO4
IO11	14	I/O/T	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5
IO12	15	I/O/T	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPIIO6
IO13	16	I/O/T	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPIIO7
IO14	17	I/O/T	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS
IO15	18	I/O/T	RTC_GPIO15, GPIO15, UORTS, ADC2_CH4, XTAL_32K_P
IO16	19	I/O/T	RTC_GPIO16, GPIO16, UOCTS, ADC2_CH5, XTAL_32K_N
IO17	20	I/O/T	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6, DAC_1
IO18	21	I/O/T	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, DAC_2, CLK_OUT3
IO19	22	I/O/T	RTC_GPIO19, GPIO19, U1RTS, ADC2_CH8, CLK_OUT2, USB_D-
IO20	23	I/O/T	RTC_GPIO20, GPIO20, U1CTS, ADC2_CH9, CLK_OUT1, USB_D+
IO21	24	I/O/T	RTC_GPIO21, GPIO21
IO26	25	I/O/T	SPICS1, GPIO26 [2]
GND	26	P	Ground
IO33	27	I/O/T	SPIIO4, GPIO33, FSPIHD
IO34	28	I/O/T	SPIIO5, GPIO34, FSPICS0
IO35	29	I/O/T	SPIIO6, GPIO35, FSPID
IO36	30	I/O/T	SPIIO7, GPIO36, FSPICLK
IO37	31	I/O/T	SPIDQS, GPIO37, FSPIQ
IO38	32	I/O/T	GPIO38, FSPIWP
IO39	33	I/O/T	MTCK, GPIO39, CLK_OUT3
IO40	34	I/O/T	MTDO, GPIO40, CLK_OUT2
IO41	35	I/O/T	MTDI, GPIO41, CLK_OUT1
IO42	36	I/O/T	MTMS, GPIO42
TXD0	37	I/O/T	U0TXD, GPIO43, CLK_OUT1
RXD0	38	I/O/T	U0RXD, GPIO44, CLK_OUT2
IO45	39	I/O/T	GPIO45
IO46	40	I	GPIO46

Figure 5. ESP32-S2-WROVER pinout [7]

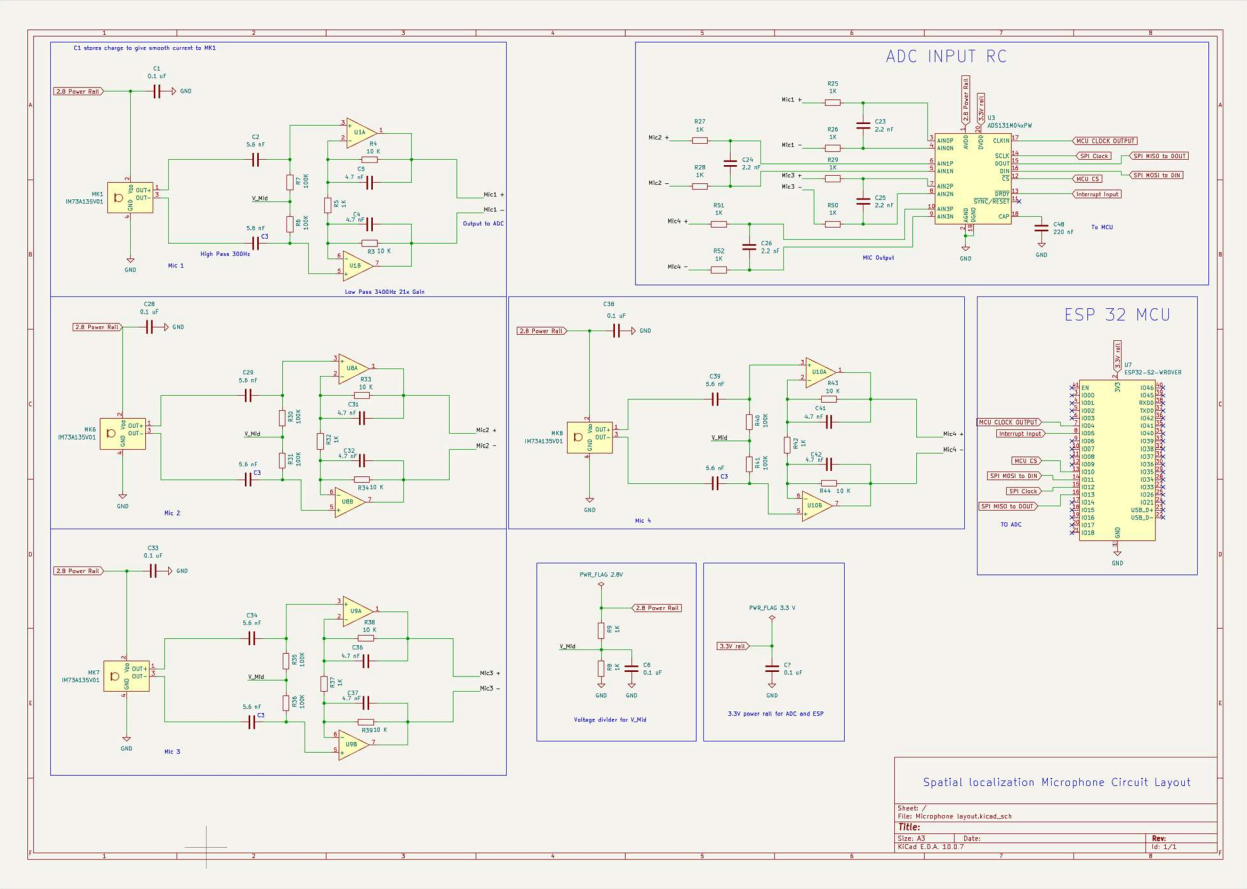


Figure 6. Full circuit schematic

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